Baron Eiley

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Professional Experience

Lockheed Martin Space - ASIC and FPGA Design Engineer **March 2023 – Present**

FPGA RTL Design and Architecture

- Designed Simulink-based models to generate synthesizable RTL signal processing modules, including interpolators, decimators, and cascaded integrator-comb (CIC) filters. Implemented and tailored MATLAB filter functions to meet specifications, extracting key parameters for HDL-compatible Simulink blocks used in SDR applications.
- Developed Verilog RTL for long-range, high-altitude vehicles to control the arming and disarming of critical hardware functions. Designs included robust finite state machines and logic to process real-time accelerometer data from ADCs.
- Validated and debugged signed fixed-point RTL for arithmetic operations processing accelerometer data using SystemVerilog testbenches, ensuring accurate FPGA-based mathematical integration to produce velocity and distance values for high-altitude interceptor vehicles.
- Architected, synthesized, and maintained FPGA-based top-level designs integrating algorithmic modules with System-on-Chip (SoC) hard processors using AMD Vivado and Microchip Libero. Designs targeted AMD Zynq Ultrascale+ and Microchip RTG4/PolarFire devices, utilizing high-performance bus protocols such as AXI and APB.
- Continuously synthesized and delivered updated FPGA bitstreams to the algorithms team for concept designs. Collected and summarized reports on timing closure and resource utilization.
- Developed C++ code for the Ultrasacale+ hard processor to interface with custom RTL designed in Vitis Model Composer for loading and unloading image data between PL and PS domains within Ultrascale FPGAs.
- Extensive practice with implementing critical ILAs for FPGA board debug and initial bring-up.

Lockheed Martin Space - Electro-Optical Engineer **August 2022 – March 2023**

C# UX/UI Firmware Development

- Developed Verilog RTL code to implement loopback for high-speed communication in electro-optic sensors using SERDES and UART protocols between FPGA and readout integrated circuits.
- Developed custom scripting language using C# to command-and-control special readout integrated circuits for highly sensitive Geiger Mode Avalanche Photo-Diode sensors (GmAPD).
- Designed UX/UI GUIs in C# to interface with equipment associated with Hardware in the Loop (HWIL) and Standardized Test Equipment (STEs).
- Wrote C# code to utilize dynamic link libraries to interface National Instruments test equipment to be controlled and monitored via custom graphical user interfaces.

Education

Master of Science in Electrical Engineering **December 2025**
San José State University, GPA: 3.44 San Jose, CA
Emphasis: Digital System Design and Artificial Intelligence

Bachelor of Science in Electronic Systems Engineering **May 2022**
California State Polytechnic University, Pomona, GPA: 3.49 Pomona, CA
Emphasis: Digital System Design, Robotics, and Control

Graduate Projects

RapidLaneFPGA: FPGA DPU Based Highway Lane Segmentation System

- Developed a MobileNetV2 lane-segmentation CNN model and deployed it to FPGA using AMD Vitis AI and PyTorch. The Project utilized Deep Processing Unit (DPU) hardware on 224×224 images.
- Achieved 77.10% DICE accuracy on FPGA vs. 77.8% software baseline with INT8 quantization at 10 FPS (~100ms latency) on AMD ZCU104 Ultrascale+ device, demonstrating near-real-time inference.

FPGA Based LeNET Neural Network for Digit Recognition

- Implemented LeNet-based MNIST digit classifier using Vitis High Level Synthesis and deployed FPGA.
- Hardware inference output validated against software golden data and validated FPGA inference on PYNQ-Z1, successfully classifying digits 0–9.

Swimmers Pose Estimation Using Computer Vision and AI/ML

- Developed a computer vision pipeline for swimmer pose estimation using MediaPipe object detection, Segment Anything Model, and YOLOv7 to isolate swimmers in image and generated coordinates to train on a custom neural network.

FPGA Based Flight Controller, Search and Rescue Drone

- Developed an FPGA-based flight controller for a six-propeller search-and-rescue drone. The drone utilized a Zynq-7000 FPGA/SoC to process RC inputs and generate PWM motor control signals within the FPGA fabric.
- Designed a power distribution board for onboard camera and lighting systems.

Technical Skills

Vivado & Vitis High Level Synthesis (HLS), Verilog, SystemVerilog, VHDL, C++ Hardware Design
MATLAB/SIMULINK, Signal Processing, Control Systems Modeling, and Hardware Arithmetic Validation
QuestaSim, FPGA Hardware Simulation & Validation
Software Programming Languages, Working knowledge of Python and C/C++
Microsoft Excel, Word, & PowerPoint